Statistics for Data Science -1

Revision: Random variables and Probability distributions

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Review of random variable, pmf, cdf, expectation, variance

- Probability mass function, Cumulative distribution function
- Expectation of a random variable
- Variance of a random variable

Some discrete distributions

- The Binomial Experiment
- Hypergeometric distribution
- Binomial versus Hypergeometric distribution
- Poisson distribution

Continuous distributions

- Uniform distribution
- Triangular distribution
- Exponential distribution
- Normal distribution

$S = \{1, 2, 3\}$

$$S = \left\{ \begin{array}{ccc} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array} \right\}$$

Random Variable

- When a probability experiment is performed, often we are not interested in all the details of the experimental result, but rather are interested in the value of some numerical quantity determined by the result.
- For example, in rolling a dice twice, often we care about only their sum of outcomes and are not concerned about the values on the individual dice.
  - That is, we may be interested in knowing that the sum is 7 and may not be concerned over whether the actual outcome was $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2)$, or $(6, 1)$.
- These quantities of interest, or, more formally, these real-valued functions defined on the sample space, are known as random variables.
- Because the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.

Rolling a dice: Sample space

X sum of outcomes
X 2 3 4 . . 12
Y Greater of outcomes
Y = 1 2 3 4 5 6

- Experiment: Roll a dice twice
- The sample space for this experiment is

$$S = \left\{ \begin{array}{l} (1,1),(1,2),(1,3),(1,4),(1,5),(1,6), \\ (2,1),(2,2),(2,3),(2,4),(2,5),(2,6), \\ (3,1),(3,2),(3,3),(3,4),(3,5),(3,6), \\ (4,1),(4,2),(4,3),(4,4),(4,5),(4,6), \\ (5,1),(5,2),(5,3),(5,4),(5,5),(5,6), \\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6), \end{array} \right\}$$

absolute
Z:
-5 -4
0 1 2 3 4 5
5

- Consider the probabilities associated with the two questions
  1. Of the outcomes, how many outcomes will result in a sum of outcomes as 7?
  2. Of the outcomes, how many outcomes will have the smaller of the outcomes as 3?
- Notice, the experiment and sample space used to answer both the questions are the same.

"DISTRIBUTION"

- Let X denote the sum of outcomes of the two rolls.
- Let Y denote the lesser of the two outcomes. If the outcomes are the same, the value of the outcome is taken as value of Y.

Outcome	X	Y	Outcome	X	Y	Outcome	X	Y
(1,1)	2	1	(3,1)	4	1	(5,1)	6	1
(1,2)	3	1	(3,2)	5	2	(5,2)	7	2
(1,3)	4	1	(3,3)	6	3	(5,3)	8	3
(1,4)	5	1	(3,4)	7	3	(5,4)	9	4
(1,5)	6	1	(3,5)	8	3	(5,5)	10	5
(1,6)	7	1	(3,6)	9	3	(5,6)	11	5
(2,1)	3	1	(4,1)	5	1	(6,1)	7	1
(2,2)	4	2	(4,2)	6	2	(6,2)	8	2
(2,3)	5	2	(4,3)	7	3	(6,3)	9	3
(2,4)	6	2	(4,4)	8	4	(6,4)	10	4
(2,5)	7	2	(4,5)	9	4	(6,5)	11	5
(2,6)	8	2	(4,6)	10	4	(6,6)	12	6

Sum of rolls of the dice

- Let X denote the sum of outcomes of the two rolls.
- X takes the values 2,3,4,5,6,7,8,9,10,11, and 12.

Value of X	Relevant event
2	$\{(1,1)\}$
3	$\{(1,2),(2,1)\}$
4	$\{(1,3),(2,2),(3,1)\}$
5	$\{(1,4),(2,3),(3,2),(4,1)\}$
$\vdots$	$\vdots$
9	$\{(3,6),(4,5),(5,4),(6,3)\}$
10	$\{(4,6),(5,5),(6,4)\}$
11	$\{(5,6),(6,5)\}$
12	$\{(6,6)\}$

- We say X is a random variable taking on one of the values 2,3,4,5,6,7,8,9,10,11, and 12 with respective probabilities

Probability of X	Probability of relevant event	Probability
$P\{X=2\}$	$P(\{(1,1)\})$	$\frac{1}{36}$
$P\{X=3\}$	$P(\{(1,2),(2,1)\})$	$\frac{2}{36}$
$P\{X=4\}$	$P(\{(1,3),(2,2),(3,1)\})$	$\frac{3}{36}$
$P\{X=5\}$	$P(\{(1,4),(2,3),(3,2),(4,1)\})$	$\frac{4}{36}$
$\vdots$	$\vdots$	$\vdots$
$P\{X=9\}$	$P(\{(3,6),(4,5),(5,4),(6,3)\})$	$\frac{4}{36}$
$P\{X=10\}$	$P(\{(4,6),(5,5),(6,4)\})$	$\frac{3}{36}$
$P\{X=11\}$	$P(\{(5,6),(6,5)\})$	$\frac{2}{36}$
$P\{X=12\}$	$P(\{(6,6)\})$	$\frac{1}{36}$

Lesser of the two values

- Let Y denote the lesser of the two outcomes. If the outcomes are the same, the value of the out come is taken as value of Y.
- Y takes the values 1,2,3,4,5, and 6.

Y	Relevant event
1	$\{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(3,1),(4,1),(5,1),(6,1)\}$
2	$\{(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(4,2),(5,2),(6,2)\}$
3	$\{(3,3),(3,4),(3,5),(3,6),(4,3),(5,3),(6,3)\}$
4	$\{(4,4),(4,5),(4,6),(5,4),(6,4)\}$
5	$\{(5,5),(5,6),(6,5)\}$
6	$\{(6,6)\}$

- We say Y is a random variable taking on one of the values 1,2,3,4,5, and 6 with respective probabilities

Y	Relevant event	Probability
1	$\{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(3,1),(4,1),(5,1),(6,1)\}$	$\frac{11}{36}$
2	$\{(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(4,2),(5,2),(6,2)\}$	$\frac{9}{36}$
3	$\{(3,3),(3,4),(3,5),(3,6),(4,3),(5,3),(6,3)\}$	$\frac{7}{36}$
4	$\{(4,4),(4,5),(4,6),(5,4),(6,4)\}$	$\frac{5}{36}$
5	$\{(5,5),(5,6),(6,5)\}$	$\frac{3}{36}$
6	$\{(6,6)\}$	$\frac{1}{36}$

Discrete and Continuous random variables

$X \quad x_1 \quad x_2 \quad \cdot \quad \cdot \quad x_n$

$x_1 \quad x_2 \quad \cdot \quad - \quad - \quad \cdot$

Definition

A random variable that can take on at most a countable number of possible values is said to be a discrete random variable.

- Thus, any random variable that can take on only a finite number or countably infinite number of different values is discrete.
- There also exist random variables whose set of possible values is uncountable.

$S =$

Definition

When outcomes for random event are numerical, but cannot be counted and are infinitely divisible, we have continuous random variables.

Discrete and continuous random variable

- A discrete random variable is one that has possible values that are discrete points along the real number line.
  - Discrete random variables typically involve counting.
- A continuous random variable is one that has possible values that form an interval along the real number line.
  - Continuous random variables typically involve measuring.

Discrete and continuous- examples

- Discrete:
  - Number of people in a household
  - Number of languages a person can speak
  - Number of times a person takes a particular test before qualifying.
  - Number of accidents in an intersection.
  - Number of spelling mistakes in a report.
- Continuous:
  - Temperature of a person.
  - Height of a person.
  - Speed of a vehicle.
  - Time taken by a person to write an exam.

Probability mass function (p.m.f)

- A random variable that can take on at most a countable number of possible values is said to be a discrete random variable.
- Let X be a discrete random variable, and suppose that it has n possible values, which we will label $x_1, x_2, \ldots, x_n$.
- For a discrete random variable X, we define the probability mass function $p(x)$ of X by

$$p(x_i) = P(X = x_i)$$

- Represent it in tabular form

X	x_1	x_2	x_3	$\ldots$	$\ldots$	x_n
$P(X = x_i)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$\ldots$	$\ldots$	$p(x_n)$

Properties of p.m.f

- The probability mass function $p(x)$ is positive for at most a countable number of values of x. That is, if X must assume one of the values $x_1, x_2, \ldots,$ then
  1. $p(x_i) \geq 0, i = 1, 2, \ldots$
  2. $p(x) = 0$ for all other values of x
- Represent it in tabular form

X	x_1	x_2	x_3			
$P(X = x_i)$	$p(x_1)$	$p(x_2)$	$p(x_3)$			

- Since X must take one of the values x_i, we have

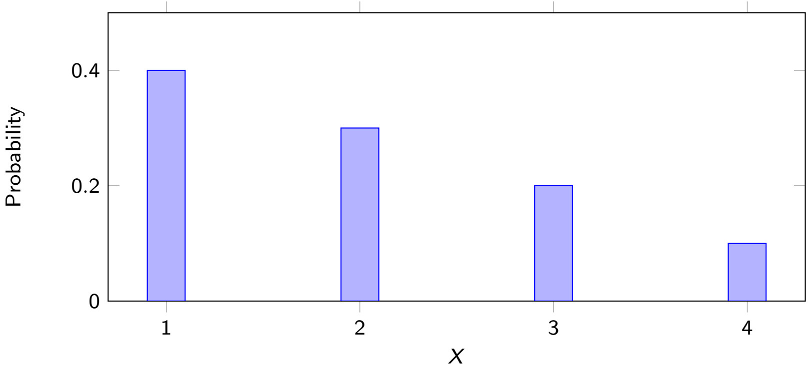
$$\sum_{i=1}^{\infty} p(x_i) = 1$$

Graph of probability mass function

- It is helpful to illustrate the probability mass function in a graphical format by plotting $P(X = x_i)$ on the y-axis against x_i on the x-axis.

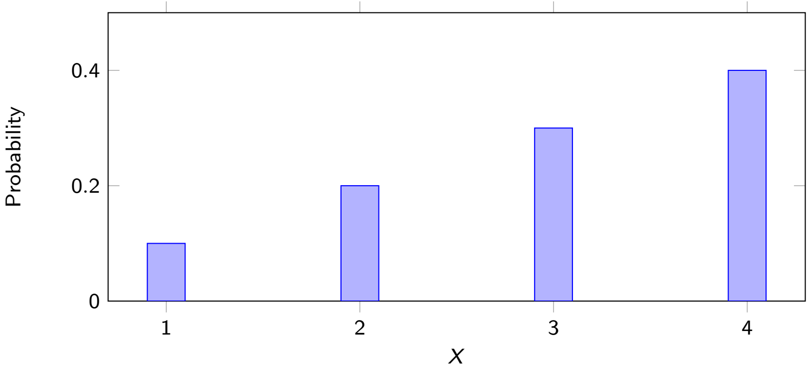
Example: positive skewed distribution

X	1	2	3	4
$P(X = x_i)$	0.4	0.3	0.2	0.1



Example: negative skewed distribution

X	1	2	3	4
$P(X = x_i)$	0.1	0.2	0.3	0.4



Cumulative distribution function

- The cumulative distribution function (cdf), F, can be expressed by
$$F(a) = P(X \leq a)$$
- If X is a discrete random variable whose possible values are $x_1, x_2, x_3, \ldots,$ where $x_1 < x_2 < x_3 \ldots$, then the distribution function F of X is a step function.

Step function

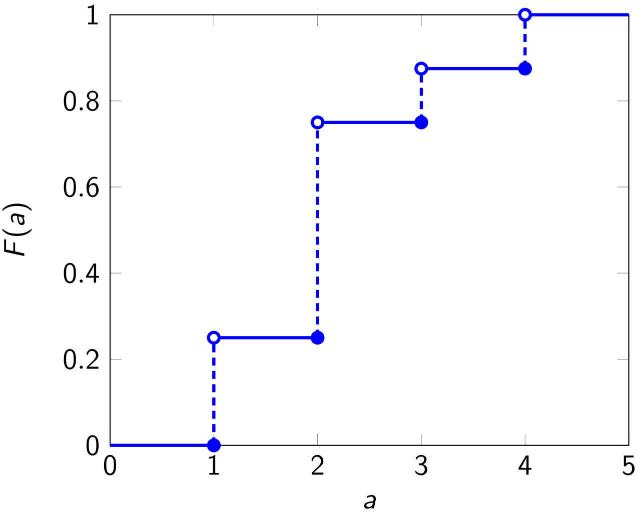
- Let X be a discrete random variable with the following probability mass function.

X	1	2	3	4
$P(X = x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$

- The cumulative distribution function of X is given by

$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{4} & 1 \leq a < 2 \\ \frac{3}{4} & 2 \leq a < 3 \\ \frac{7}{8} & 3 \leq a < 4 \\ 1 & 4 \leq a \end{cases}$$

Note that the size of the step at any of the values 1, 2, 3, and 4 is equal to the probability that X assumes that particular value.


1
0.8
0.6
0.4
0.2
0
$F(a)$
0
1
2
3
4
5
a

Expectation of a random variable

Definition

Let X be a discrete random variable taking values $x_1, x_2, \ldots,$. The expected value of X denoted by $E(X)$ and referred to as Expectation of X is given by

$$E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i)$$

- The Expectation of a random variable can be considered the "long-run-average" value of the random variable in repeated independent observations.
- Lets apply the definition to the examples we have considered before

Introduction

- The expected value of a random variable gives the weighted average of the possible values of the random variable, it does not tell us anything about the variation, or spread, of these values.
- For instance, consider random variables X, Y, and Z, whose values and probabilities are as follows:
  - $X = 0$ with probability 1
  - $Y = \begin{cases} -2 & \text{with probability} \frac{1}{2} \\ 2 & \text{with probability} \frac{1}{2} \end{cases}$
  - $Z = \begin{cases} -20 & \text{with probability} \frac{1}{2} \\ 20 & \text{with probability} \frac{1}{2} \end{cases}$
- $E(X) = E(Y) = E(Z) = 0$. However, we notice spread of Z is greater than spread of Y which is greater than spread of X
- Need for a measure of spread.

Variance of a random variable

- Let's denote expected value of a random variable X by the Greek alphabet μ.

Definition

Let X be a random variable with expected value μ, then the variance of X, denoted by $Var(X)$ or $V(X)$, is defined by

$$Var(X) = E(X - \mu)^2$$

- In other words, the Variance of a random variable X measures the square of the difference of the random variable from its mean, μ, on the average.

Computational formula for $Var(X)$

$V(X) = E(X-\mu)^2$

$E(X^2 + \mu^2 - 2X\mu)$

$E(X^2) + \mu^2 - 2\mu E(X)$

$E(X^2) + \mu^2 - 2\mu^2$

- $Var(X) = E(X-\mu)^2 =$
- $(X-\mu)^2 = X^2 - 2X\mu + \mu^2$
- Using properties of expectation we know $E(X^2 - 2X\mu + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2$ which is same as $E(X^2) - \mu^2$
- Let's compute the variance of the random variables discussed earlier.

$V(X) = E(X^2) - (E(X))^2$

X	-1	0	1
	$1/3$	$1/3$	$1/3$
X^2	$1/3$	0	1 / $2/3$

$E(X) = 0 \quad E(X^2) = 2/3$

Bernoulli random variable

- A random variable that takes on either the value 1 or 0 is called a Bernoulli random variable.
- Let X be a Bernoulli random variable that takes on the value 1 with probability p.
- The probability distribution of the random variable is

X	0	1
$P(X = x_i)$	$1 - p$	p

- Expected value of a Bernoulli random varaible:

$$E(X) = 0 \times (1 - p) + 1 \times p = p$$

Discrete uniform random variable

- Let X be a random variable that is equally likely to takes any of the values $1, 2, \ldots, n$
- Probability mass function

X	1	2	...	n
$P(X = x_i)$	$\frac{1}{n}$	$\frac{1}{n}$	...	$\frac{1}{n}$

- $E(X) = \sum_{i=1}^{n} x_i p(x_i) = \frac{(1 \times 1) + (2 \times 1) + \ldots + (n \times 1)}{n} = \frac{n(n+1)}{2 \times n} = \frac{(n+1)}{2}$

Independent and identically distributed Bernoulli trials

- A collection of Bernoulli trials defines iid Bernoulli random variables, one for each trial.
- The abbreviation iid stands for independent and identically distributed.

Definition

A collection of random variables is iid if the random variables are independent and share a common probability distribution

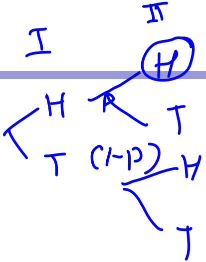
1/2 1/2

Binomial random variable

iid

Tossing

2



- Suppose that n independent trials are performed, each of which results in either a "success" with probability p or a "failure" with probability $1 - p$.
- Let X is the total number of successes that occur in n trials, then X is said to be a binomial random variable with parameters n and p.

n independent trials, $X =$ number of successes

- Let there be n independent Bernoulli trials.
- Let p is probability of success.
- $X=$ number of succeses in n independent trials.
- The probabilities of outcomes of the independent trials are

S.No	Outcome	Number of successes	Probabilities
1	(s,s,...,s)	n	$p \times p \times \ldots \times p$
2	(s,s,...,f)	$n-1$	$p \times p \times \ldots \times (1-p)$
3	(s,...,f,s)	$n-1$	$p \ldots \times p \times (1-p) \times p$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
2^{n-2}	(f,...,s,f)	1	$(1-p) \times (1-p) \ldots \times p \times (1-p)$
2^{n-1}	(f,f,...,s)	1	$(1-p) \times (1-p) \ldots \times p$
2^n	(f,f,...,f)	0	$(1-p) \times (1-p) \ldots \times (1-p)$

n independent trials, X = number of successes

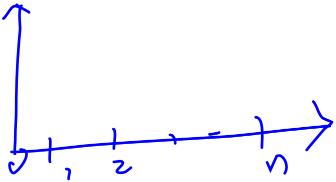
- Consider any outcome that results in a total of i successes.
  - This outcome will have a total of i successes and $(n-i)$ failures.
  - Probability of i success and $(n-i)$ failures $= p^i \times (1-p)^{(n-i)}$

.

- There number of different outcomes that result in i successes and $(n-i)$ failures$=\binom{n}{i}$
- The probability of i successes in n trials is given by

$$P(X=i) = \binom{n}{i} \times p^i \times (1-p)^{(n-i)}$$

Binomial random variable



Definition

X is a binomial random variable with parameters n and p that represents the number of successes in n independent Bernoulli trials, when each trial is a success with probability p. X takes values $0, 1, 2, \ldots, n$ with the probability

$$P(X = i) = \binom{n}{i} \times p^i \times (1 - p)^{(n-i)}$$

Shape of the pmf for same n different p

A binomial distribution is

- right skewed if $p < 0.5$
- is symmetric if $p = 0.5$
- is left skewed if $p > 0.5$

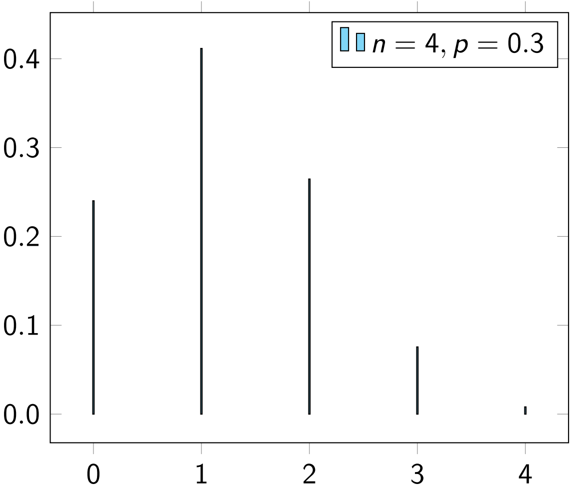
. We demonstrate the same for $n = 4$ and different p

$n = 4, p = 0.3, X =$ number of successes

- Let $n = 4$ independent Bernoulli trials.
- Let $p = 0.3$ is probability of success.
- Let $X =$ number of successes in 4 independent trials.
- The probability distribution of X

X	0	1	2	3	4
$P(X = i)$	0.2401	0.4116	0.2646	0.0756	0.0081

Graph of pmf of Binomial distribution- Right skewed

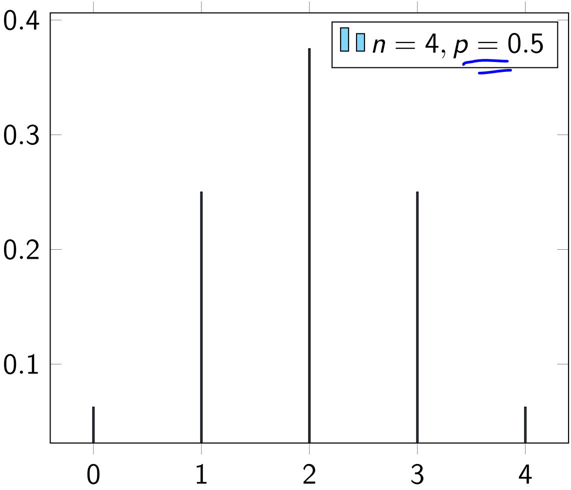


$n = 4, p = 0.5, X =$ number of successes

- Let $n = 4$ independent Bernoulli trials.
- Let $p = 0.5$ is probability of success.
- Let $X =$ number of successes in 4 independent trials.
- The probability distribution of X

X	0	1	2	3	4
$P(X = i)$	0.0625	0.25	0.375	0.25	0.0625

Graph of pmf of Binomial distribution- symmetric



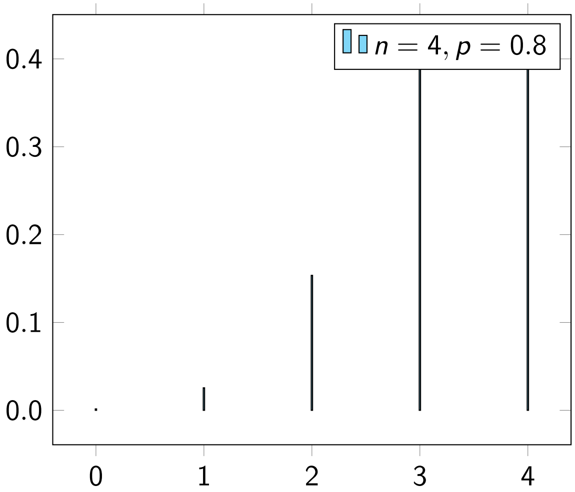
$X = 0 \quad 1 \quad 2 \quad 3 \quad 4$

$n = 4, p = 0.8, X =$ number of successes

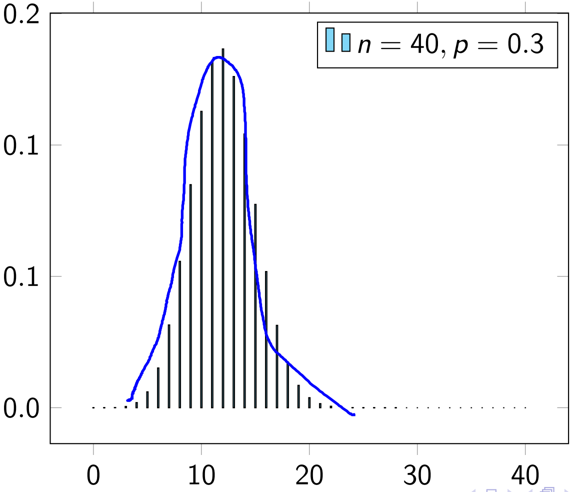
- Let $n = 4$ independent Bernoulli trials.
- Let $p = 0.8$ is probability of success.
- Let $X =$ number of successes in 4 independent trials.
- The probability distribution of X

X	0	1	2	3	4
$P(X = i)$	0.0016	0.0256	0.1536	0.4096	0.4096

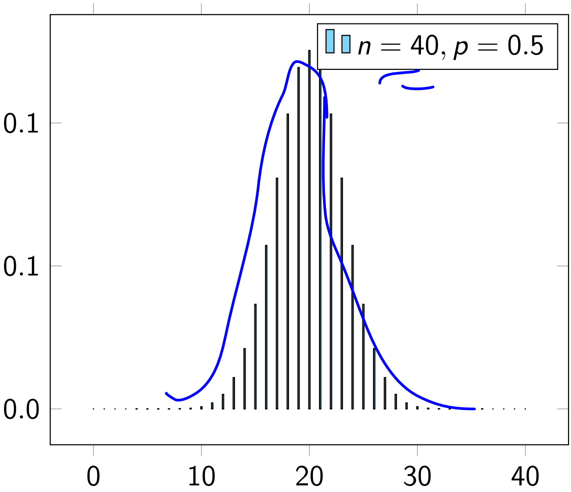
Graph of pmf of Binomial distribution- left skewed



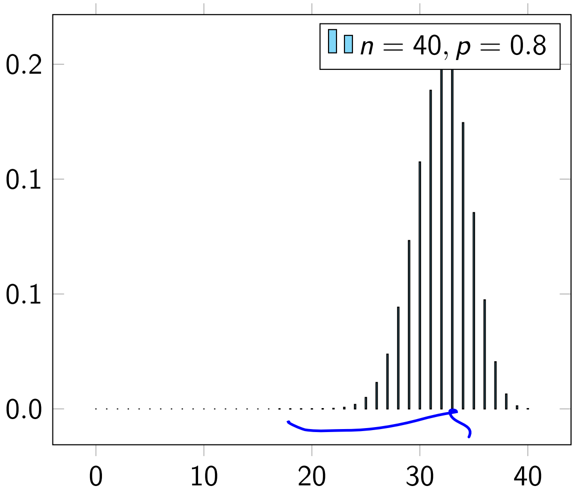
Graph of pmf of Binomial distribution- Right skewed- large n



Graph of pmf of Binomial distribution- symmetric-large n



Graph of pmf of Binomial distribution- left skewed- large n



Effect of n and p on shape of distribution

- small n, small p- right skewed
- small n, large p- left skewed
- small n $p = 0.5$- symmetric
- For large n, the binomial distribution approaches symmetry.

Expectation and Variance of Binomial Random Variable

- A binomial random variable $X \sim Bin(n, p)$ is equal to the number of successes in n independent trials when each trial is a success with probability p.
- We can represent X as

$$X = X_1 + X_2 + \ldots + X_n$$

  where X_i is equal to 1 if trial i is a success and is equal to 0 if trial i is a failure.
- $P(X_i = 1) = p$ and $\mathrm{P}(X_i = 0) = (1 - p)$

Expectation and Variance of Binomial Random Variable

- $X = X_1 + X_2 + \ldots + X_n$
- $E(X) = E(X_1 + X_2 + \ldots + X_n) = E(X_1) + E(X_2) + \ldots + E(X_n)$
  - $E(X) = p + p + \ldots + p = np$
- $V(X) = V(X_1 + X_2 + \ldots + X_n) = V(X_1) + V(X_2) + \ldots + V(X_n)$
  - $V(X) = p(1-p) + p(1-p) + \ldots + p(1-p) = np(1-p)$

Result

Using the fact that the expectation of the sum of random variables is equal to the sum of their expectations, we see that Expectation of a Binomial random variable X is

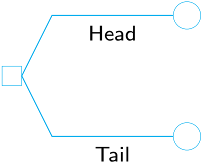
$$E(X) = np$$

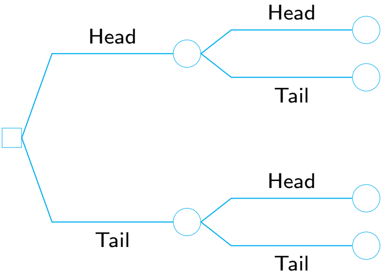
Also, since the variance of the sum of independent random variables is equal to the sum of their variances, we have variance of a Binomial random variable is

$$Var(X) = np(1-p)$$

Limitations of Binomial distribution

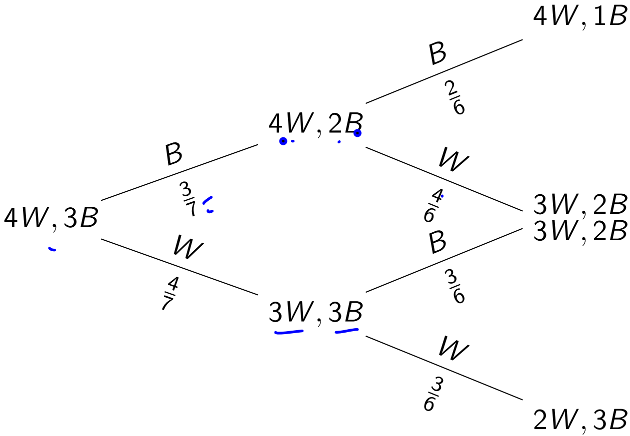
- Suppose we are interested in finding the probability of success in n trials, where the trials are independent.





Limitations of Binomial distribution

- Suppose trials are not independent and the probability of "success" is not the same for all trials.



Introduction

For the hypergeometric to work,

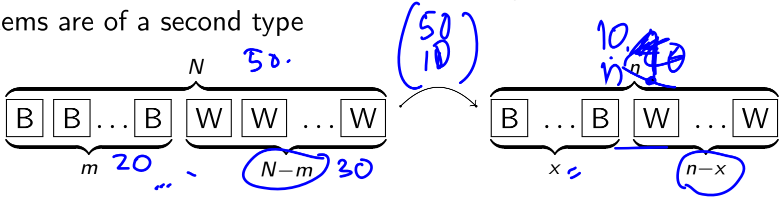
- The population must be dividable into two and only two independent subsets (black balls and white balls in our example).
- The experiment must have changing probabilities of success with each experiment (the fact that balls are not replaced after the draw in our example makes this true in this case).
- Another way to say this is that you sample without replacement and therefore each draw is not independent.

The Hypergeometric distribution

- A discrete random variable (RV) that is characterized by:
  - A fixed number of trials.
  - The probability of success is not the same from trial to trial.
- We sample from two groups of items when we are interested in only one group.
- X is defined as the number of successes out of the total number of items chosen.

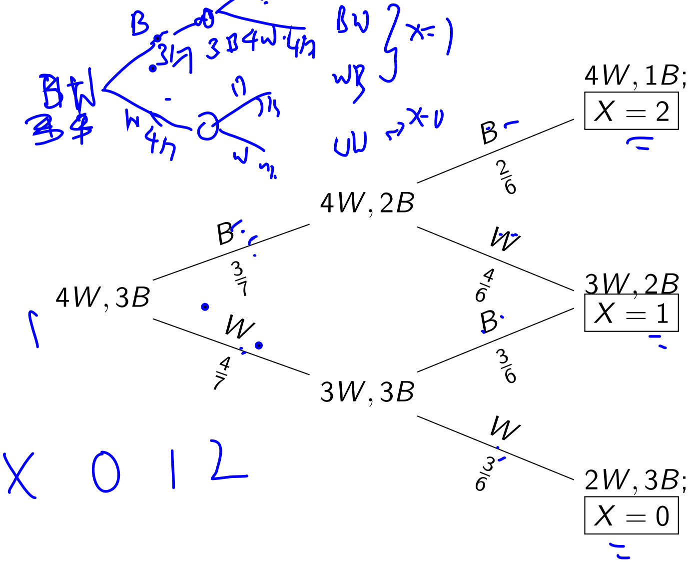
Understanding the Hypergeometric distribution

If we randomly select n items without replacement from a set of N items of which: m of the items are of one type and $N-m$ of the items are of a second type



Let X be the number of items of type 1, then the probability mass function of the discrete random variable, X, is called the hypergeometric distribution and is of the form:

$$P(X=x)=\frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}};\ x=0,1,\ldots,n$$


4W, 3B
B
3/7
W
4/7
4W, 2B
3W, 3B
B
2/6
W
4/6
B
3/6
W
3/6
4W, 1B;
X = 2
3W, 2B
X = 1
2W, 3B;
X = 0
BB → X=2
BW
WB
X=1
WW → X=0
X 0 1 2

Examples: Choosing balls without replacement

A bag consists of 7 balls of which 4 are white and 3 are black. A student randomly samples two balls without replacement. Let X be the number of black balls selected.

- Here, $N = 7, n = 2, m = 3$
- X takes values: $0, 1, 2$
- $P(X = i) = \frac{\binom{3}{i}\binom{4}{2-i}}{\binom{7}{2}}; i = 0, 1, 2$
- The pmf

i	0	1	2
P(X=i)	$\frac{12}{42}$	$\frac{24}{42}$	$\frac{6}{42}$

Handwritten notes: $\binom{m}{i}\binom{N-m}{n-i}$ over $\binom{N}{n}$

$7 \rightarrow 2$

BBB WWWW

BB
BW
WW

X= 0 1 2

Expectation and Variance (N, n, m)

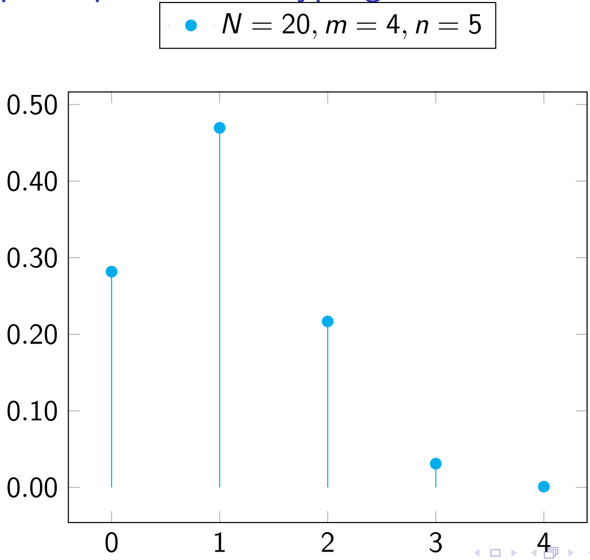
Let X follow a hypergeometric distribution in which n objects are selected from N objects with m of the objects being one type, and $N - m$ of the objects being a second type.

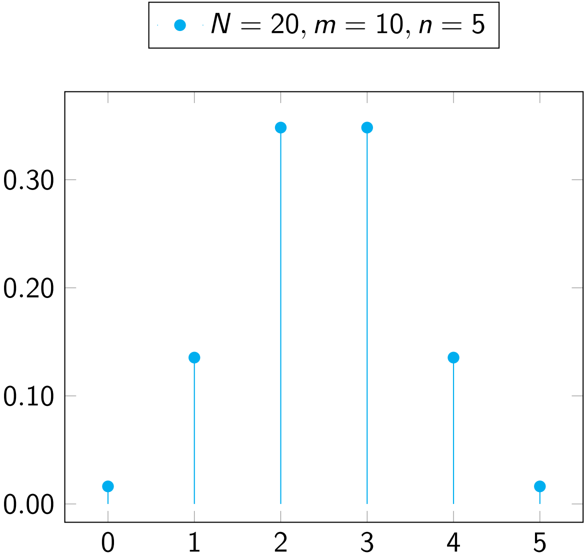
- $n\hat{p}$ $\quad E(X) = \frac{nm}{N}$ $\quad n \cdot \frac{m}{N} = \hat{p}$
- $Var(X) = n\frac{m}{N}\frac{N-m}{N}\frac{N-n}{N-1}$

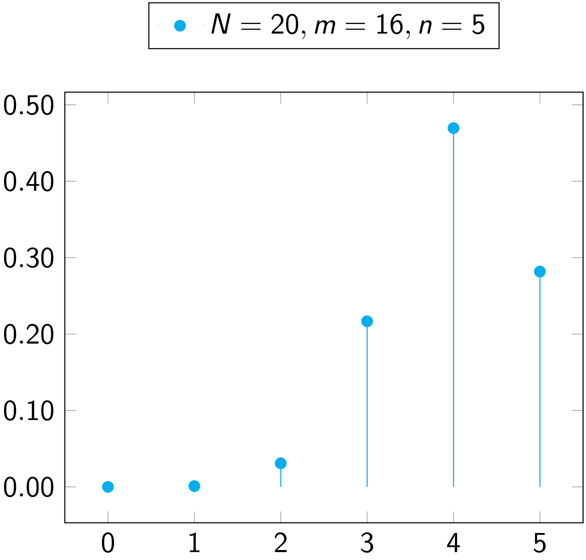
$n \cdot \frac{m}{N}\left(1 - \frac{m}{N}\right)$

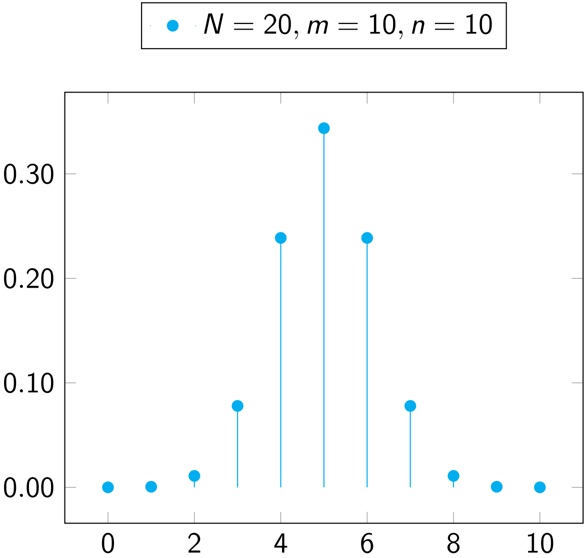
$n\hat{p}(1-\hat{p})$

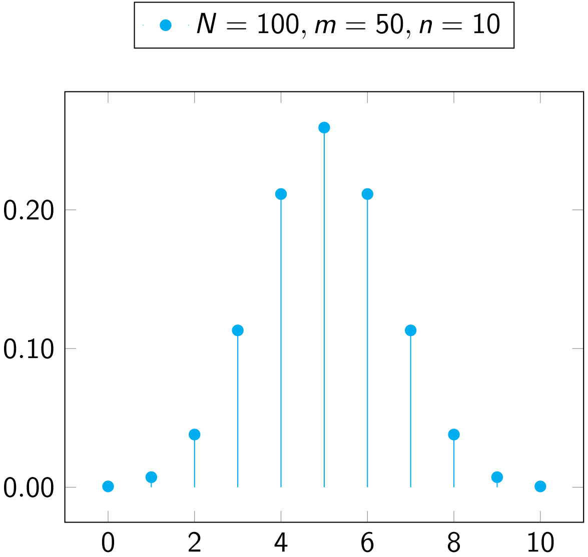
Graph of pmf of the Hypergeometric distribution

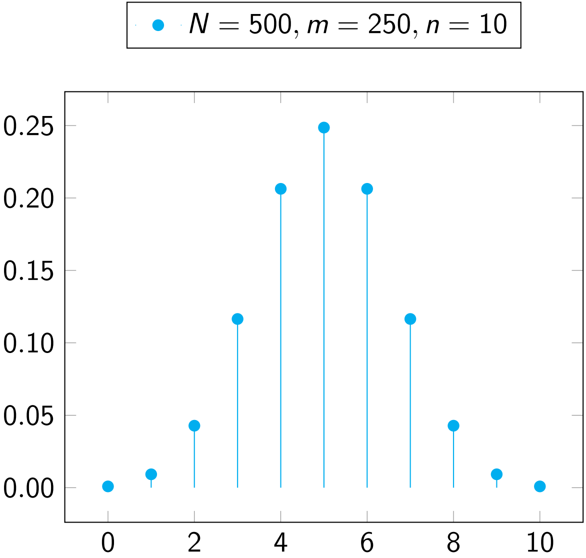



$N = 20, m = 10, n = 5$
0.30
0.20
0.10
0.00
0
1
2
3
4
5


$N = 20, m = 16, n = 5$
0.50
0.40
0.30
0.20
0.10
0.00
0
1
2
3
4
5


$N = 20, m = 10, n = 10$
0.30
0.20
0.10
0.00
0
2
4
6
8
10

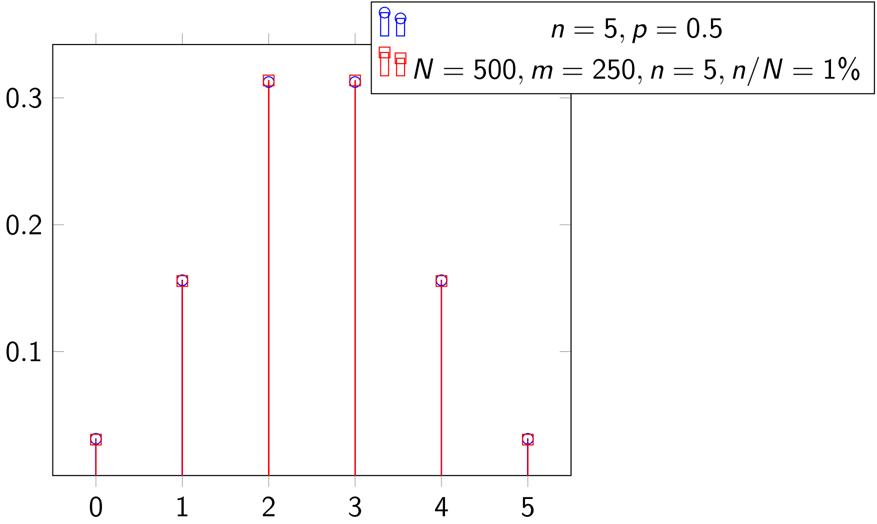

$N = 100, m = 50, n = 10$
0.20
0.10
0.00
0
2
4
6
8
10


N = 500, m = 250, n = 10
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
10

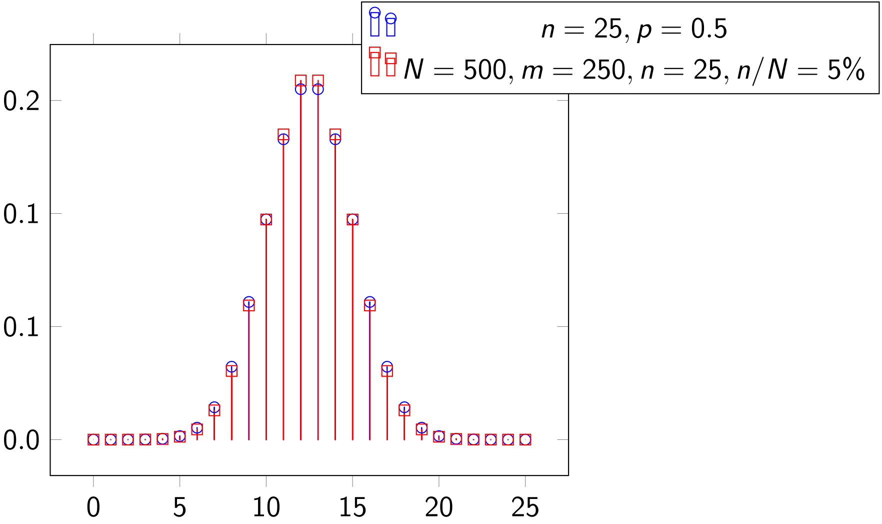
Expectation and variance

- $X \sim Hypergeometric(N, m, n)$
  - $E(X) = \dfrac{nm}{N}$
  - $Var(X) = n\frac{m}{N}\frac{N-m}{N}\frac{N-n}{N-1}$
- $Y \sim Bin\left(n, \dfrac{m}{N}\right)$
  - $E(Y) = \dfrac{nm}{N}$
  - $Var(Y) = n\frac{m}{N}\frac{N-m}{N}$
- Consider the term: $\frac{N-n}{N-1}$
  - For $n = 1$, replacement has no effect both are Bernoulli trial
  - For $n = N$, the whole population is sampled- hence variance is zero.
- If the population N is very large compared to the sample size n (i.e. $N >> n$) then Hypergeometric(N, m, n) is about Binomial$(n, \dfrac{m}{N})$.

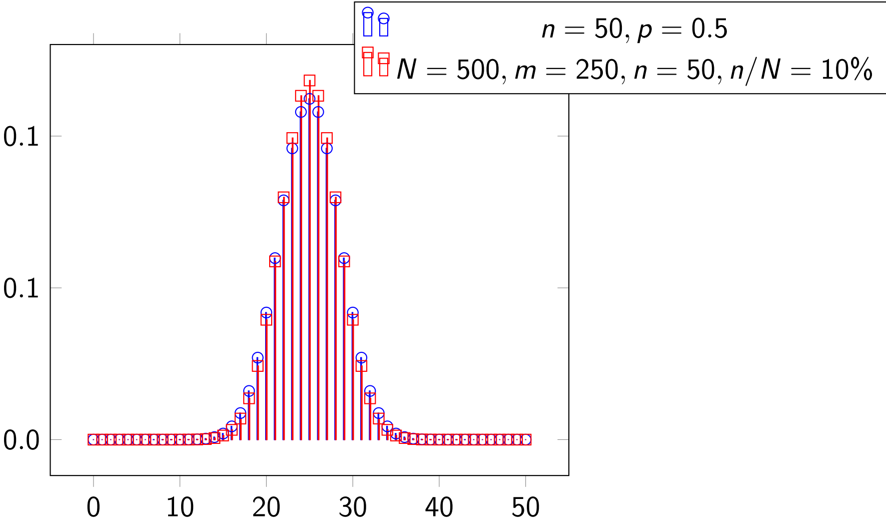
Binomial versus Hyergeometric distribution



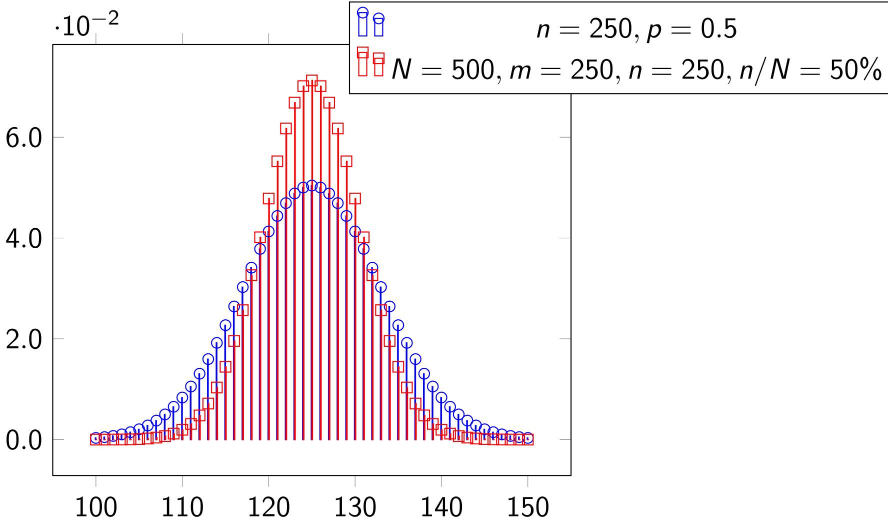
Binomial versus Hyergeometric distribution



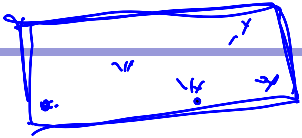
Binomial versus Hyergeometric distribution



Binomial versus Hypergeometric distribution



Introduction



- The Poisson probability distribution gives the probability of a number of events occurring in a fixed interval of **time** or **space**.
- We assume that these events happen with a known average rate, λ, and independently of the time since the last event.
- Let X denote the number of times an event occurs in an interval of time (or space).
- We say $X \sim Poisson(\lambda)$, in other words, X is a random variable that follows Poisson distribution with parameter λ.
- The Poisson distribution may be used to approximate the Binomial distribution if the probability of success is “small” and the number of trials is “large” .

Probability mass function of Poisson

The distribution, with an average number of λ events per interval, is defined as Poisson discrete random variable, $X \sim Poisson(\lambda)$, with the p.m.f given by

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, x = 0, 1, 2, \ldots$$

- The random varaible X represents the number of events per time interval (In the example: number of vehicles passing per minute)
- e is the mathematical constant 2.718

Expectation and Variance of Poisson distribution

- For a Poisson random variable $X \sim Poisson(\lambda)$, both the expected value and the variance of X are equal to λ.

Examples of Poisson distribution

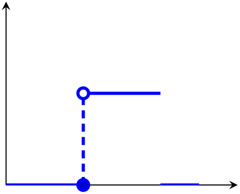
$$P(X=3) = \frac{e^{-0.5} \, 0.5^3}{3!}$$

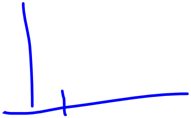
- Events occurring in fixed interval of time
  1. Number of vehicles passing through a traffic intersection in a fixed time interval of one minute.
  2. Number of people withdrawing money from a bank in a fixed time interval of fifteen minutes.
  3. Number of telephone calls received per minute at a call center
- Events occurring in fixed interval of space
  1. Number of typos (incorrect spelling) in a book.
  2. Number of defects in a wire cable of finite length.
  3. Number of defects per meter in a roll of cloth

Uniform distribution $U(a, b)$

$p(X = x_i) = 0$

- A continuous random variable has a uniform distribution, denoted $X \sim U(a, b)$,





probability density function is:

$$f(x) = \begin{cases} \frac{1}{(b-a)} & a \leqslant x \leqslant b \\ 0 & \text{otherwise} \end{cases}$$

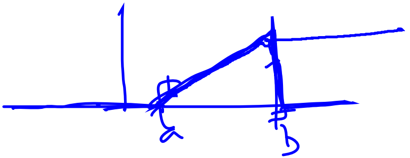
Standard uniform distribution

- A random variable has the standard uniform distribution with minimum 0 and maximum 1 if its probability density function is given by

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

  The standard uniform distribution plays an important role in random variate generation.
- Verify $f(x)$ is a pdf
  - $f(x) \geq 0$, for $0 < x < 1$
  - $\int_{-\infty}^{\infty} f(x)dx = \int_0^1 f(x)dx = 1$

Cumulative distribution of Uniform distribution



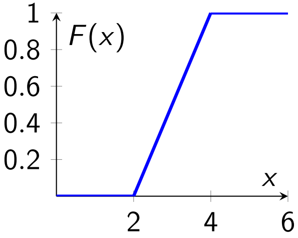
For $X \sim U(a, b)$

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b) \\ 1 & \text{for } x \geq b \end{cases}$$

$F(X) =$

$$\frac{-a}{b-a} + \frac{1}{b-a} x$$

Cumulative distribution of Uniform distribution



- When $x < a$ $F(x) = 0$
- When $x > b$ F(x) =1
- $a < x < b$ The slope of the line between and is $\frac{1}{(b-a)}$.

Expectation of $X \sim U(a, b)$

- $X \sim U(a, b)$;
$$E(X) = \frac{a+b}{2}$$
-
$$\begin{aligned} E(X) &= \int_a^b xf(x), dx \\ &= \int_a^b x \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \frac{x^2}{2} \Bigg|_a^b \\ &= \frac{b+a}{2} \end{aligned}$$

Variance of $X \sim U(a, b)$

- $X \sim U(a, b)$;
$$Var(X) = \frac{(b-a)^2}{12}$$

-
$$\begin{aligned} E(X^2) &= \int_a^b x^2 f(x), dx &= \int_a^b x^2 \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \frac{x^3}{3} \Bigg|_a^b &= \frac{b^2+a^2+ab}{3} \end{aligned}$$

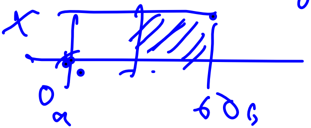
-
$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{b^2+a^2+ab}{3} - \left(\frac{b+a}{2}\right)^2 \\ &= \frac{(b-a^2)}{12} \end{aligned}$$

Example: Computing probabilities given distribution

Suppose that X is a uniform random variable over the interval $(0, 1)$. Find

1. $P(X > 1/3) = 2/3$
2. $P(X \leq 0.7) = 0.7$
3. $P(0.3 < X \leq 0.9) = 0.6$
4. $P(0.2 \leq X < 0.8) = 0.6$

Example: Application-question



$$\int_{30}^{60} \frac{1}{60}\, dx$$

You are to meet a friend at 2 p.m. However, while you are always exactly on time, your friend is always late and indeed will arrive at the meeting place at a time uniformly distributed between 2 and 3 p.m. Find the probability that you will have to wait

1. At least 30 minutes
2. Less than 15 minutes
3. Between 10 and 35 minutes
4. Less than 45 minutes

$f(x) = \frac{1}{60} \quad 0 \le x \le 60$

$P(X \ge 30)$

$P(X \le a) = \int f(x)\, dx$ $\quad \int f(x)\, dx$ $\quad \int_{10}^{35} f(x)\, dx$ $\quad \int f(x)\, dx$

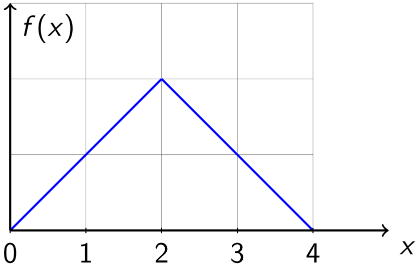
Example: Application-solution

Let X denote the amount of time you will have to wait.
$X \sim U(0, 60)$

1. At least 30 minutes $= P(X \geq 30) = 30/60 = 1/2$
2. Less than 15 minutes $= P(X < 15) = 15/60 = 1/4$
3. Between 10 and 35 minutes
   $= P(10 \leq X \leq 35) = 25/60 = 5/12$
4. Less than 45 minutes $= P(X < 45) = 45/60 = 3/4$

Triangular distribution

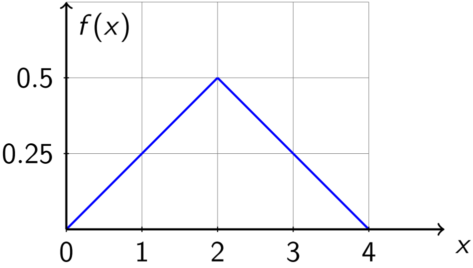
It is now 2 p.m., and Joan is planning on studying for her statistics test until 6 p.m., when she will have to go out to dinner. However, she knows that she will probably have interruptions and thinks that the amount of time she will actually spend studying in the next 4 hours is a random variable whose probability density curve is as follows:



Questions

1. What is the height of the curve at the value 2?
2. What is the probability she will study more than 3 hrs?
3. What is the probability she will study between 1 and 3 hrs?

Solution



1. What is the height of the curve at the value 2? $=1/2 unit$
2. What is the probability she will study more than 3 hrs? $=1/8$
3. What is the probability she will study between 1 and 3 hrs? $=3/4$

Exponential distribution

$$f(x) = \begin{cases} \alpha e^{-\alpha x} \\ 0 \end{cases}$$

A continuous random variable whose probability density function is given, for some λ (α) > 0, by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

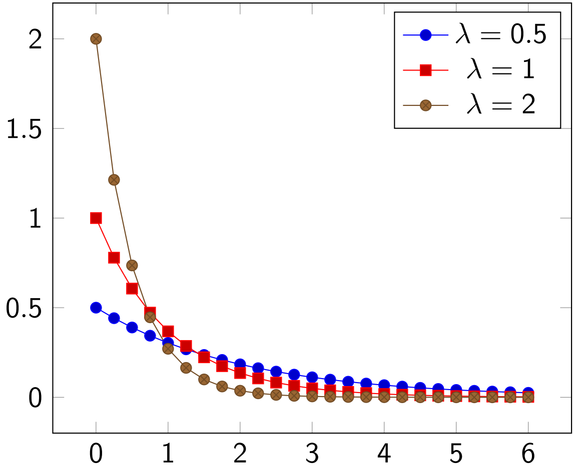
$$f(x) = \begin{cases} \alpha e^{-\alpha x} \end{cases}$$

is said to be an exponential random variable (or, more simply, is said to be exponentially distributed) with parameter λ (α).

$$F(x) = P(X \leq x) = \int_0^x \alpha e^{-\alpha x}\,dx = \alpha \int_0^x e^{-\alpha x}\,dx$$

$$F(x) = 1 - e^{-\alpha x}$$

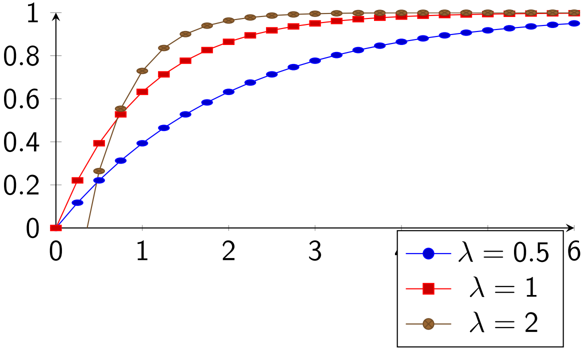
Graph of pdf for different values of λ



cdf of Exponential distribution

$$
\begin{aligned}
F(a) &= P(X \leq a) \\
&= \int_0^a \lambda e^{-\lambda x} dx \\
&= -e^{-\lambda x}\Big|_0^a \\
&= 1 - e^{-\lambda a}
\end{aligned}
$$

Graph of cdf for different values of λ



Expectation and variance of exponential distribution

$X \sim exp(\lambda)$

- $E(X) = \frac{1}{\lambda}$
- $Var(X) = \frac{1}{\lambda^2}$

- It can be shown through integration by parts

$$E(X^n) = \frac{n}{\lambda} E(X^{n-1})$$

  - $E(X) = \frac{1}{\lambda}$
  - $E(X^2) = \frac{2}{\lambda}\frac{1}{\lambda} = \frac{2}{\lambda^2}$
  - Hence $Var(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$

Thus, the mean of the exponential is the reciprocal of its parameter λ, and the variance is the mean squared.

Application

In practice, the exponential distribution often arises as the distribution of the amount of time until some specific event occurs.

- Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 0.1$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait

  P(X

  a more than 10 minutes
  b between 10 and 20 minutes.

Solution Let X denote the length of the call made by the person in the booth. $X \sim exp(0.5)$

$1 - P(X < 10)$

a more than 10 minutes $= P(X > 10) = e^{-1} \approx 0.368$

b between 10 and 20 minutes=
$P(10 < X < 20) = F(20) - F(10) = e^{-1} - e^{-2} \approx 0.233$

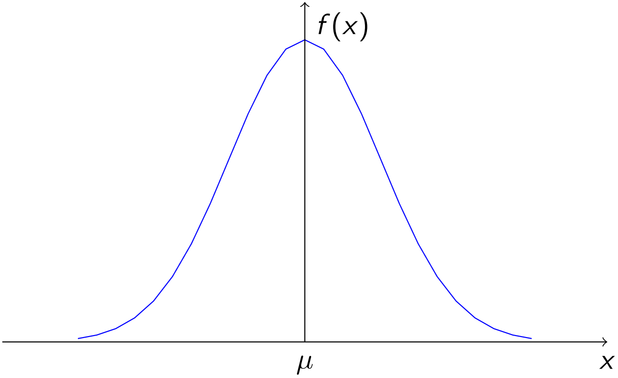
pdf of a Normal distribution

- The probability density function of a normal random variable X is determined by two parameters:
  - the expected value of X, denoted by μ
  - the variance of X, denoted by σ^2.
- We write $X \sim N(\mu, \sigma^2)$
- The Density function is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty$$

Shape of Normal pdf

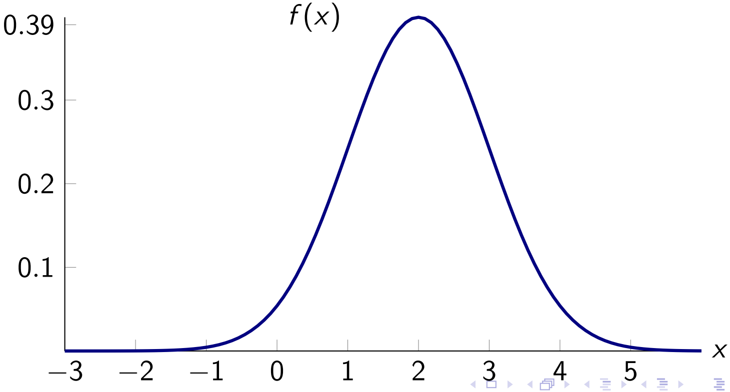
- The normal probability density function is a bell-shaped density curve that is symmetric about the value μ.
- Its variability is measured by standard deviation σ .



Height of curve

The height of the curve above point x on the abscissa is

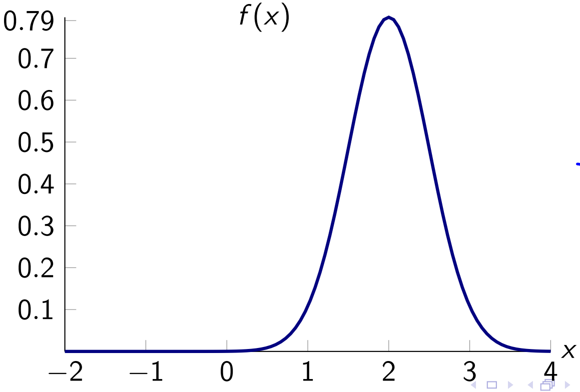
$$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; \mu = 2, \sigma = 1, \ height = 0.399$$



Height of curve

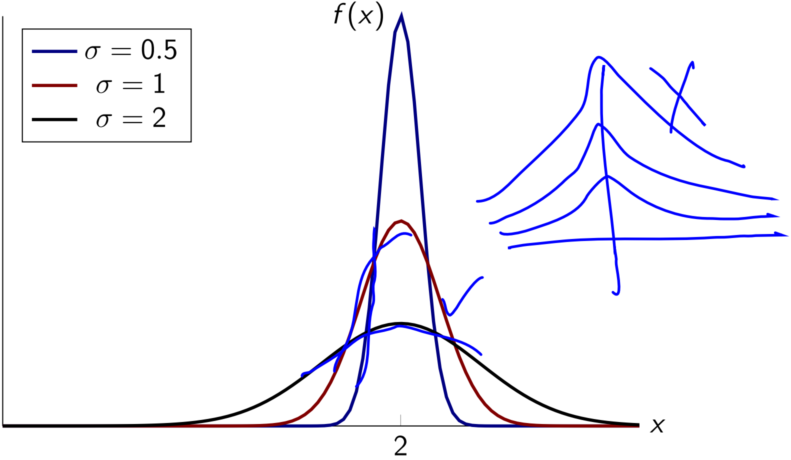
The height of the curve above point x on the abscissa is

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; \mu = 2, \sigma = 0.5, \ \textit{height} = 0.798$$



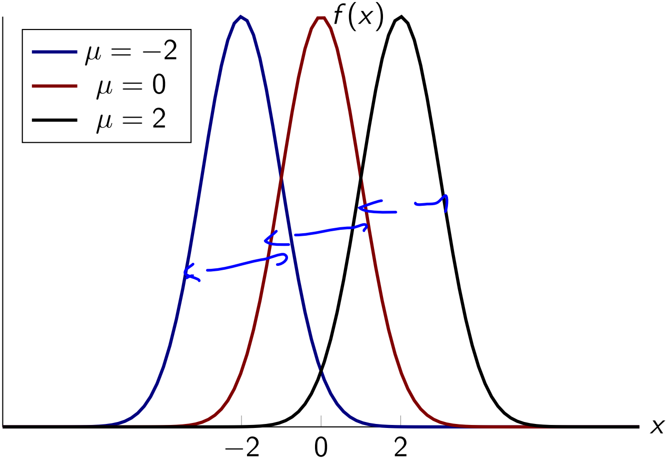
$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Same μ different σ



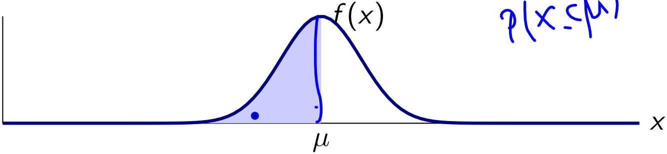
The larger the σ, the flatter and more spread out is the distribution.

Same σ different μ

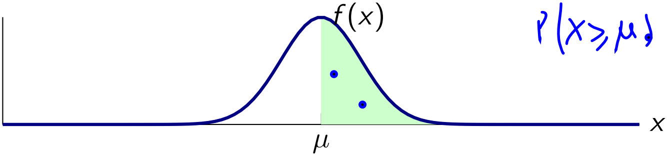


Symmetric around mean

The probability density function of a normal random variable X is symmetric about its expected value μ, it follows that X is equally likely to be on either side of μ. That is,



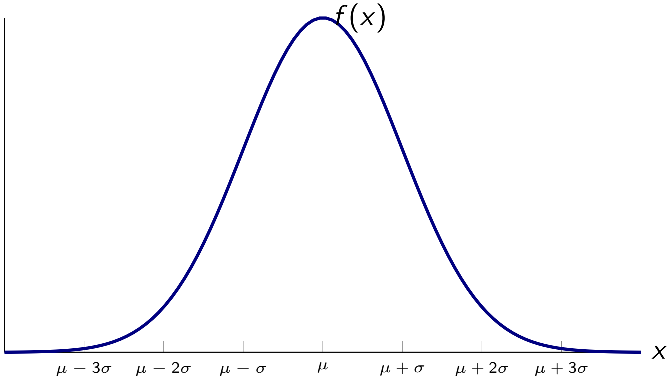
$P(X \leq \mu)$



$P(X \geq \mu)$

$$P(X \leq \mu) = P(X \geq \mu) = \frac{1}{2}$$

General μ and σ

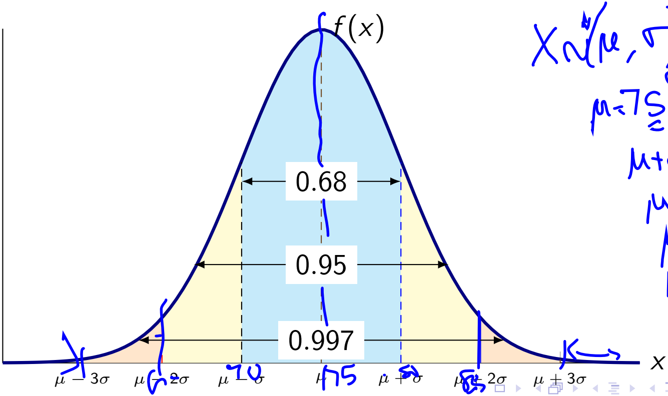


1. bell shaped,
2. centered at μ and
3. close to the horizontal axis outside the range from $\mu - 3\sigma$ to $\mu + 3\sigma$

Approximation rule

A normal random variable with mean μ and standard deviation σ will be

- Between $\mu - \sigma$ and $\mu + \sigma$ with approx. probability 0.68
- Between $\mu - 2\sigma$ and $\mu + 2\sigma$ with approx. probability 0.95
- Between $\mu - 3\sigma$ and $\mu + 3\sigma$ with approx. probability 0.997



$X \sim N(\mu, \sigma)$

$\mu = 75$, $\sigma = 5$

$\mu + \sigma = 80$

$\mu + 2\sigma = 85$

$\mu + 3\sigma = 90$

$\mu - \sigma = 70$

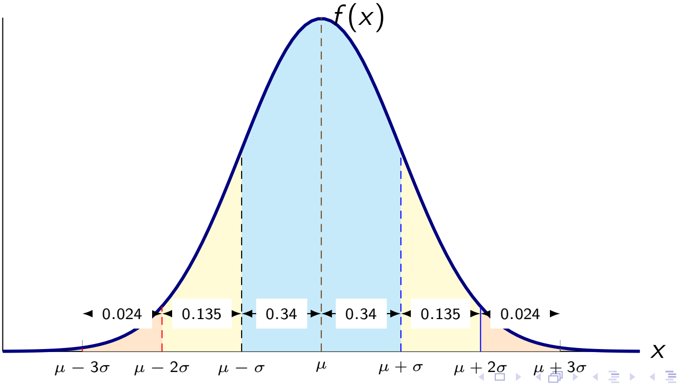
$\mu - 2\sigma = 65$

$\mu - 3\sigma = 60$

Approximation rule

A normal random variable with mean μ and standard deviation σ will be

- Between $\mu - \sigma$ and $\mu + \sigma$ with approx. probability 0.68
- Between $\mu - 2\sigma$ and $\mu + 2\sigma$ with approx. probability 0.95
- Between $\mu - 3\sigma$ and $\mu + 3\sigma$ with approx. probability 0.997

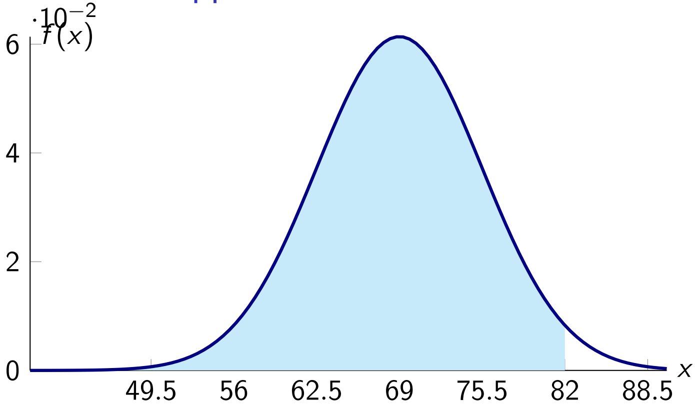


Application of approximation rule

The heights of a certain population of males are normally distributed with mean 69 inches and standard deviation 6.5 inches. Approximate the proportion of this population whose height is less than 82 inches.[1]

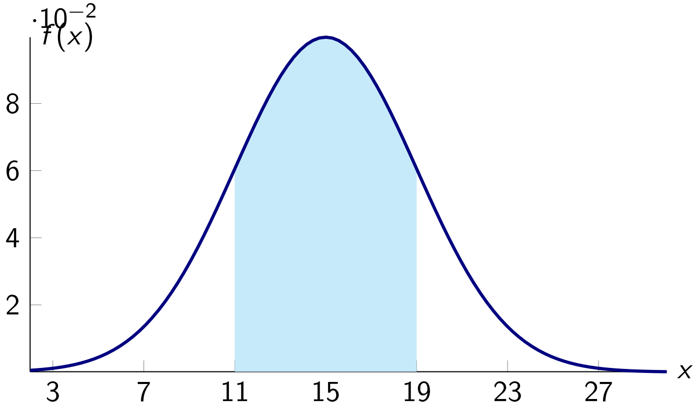
[1] Ross, Sheldon M. Introductory statistics. Academic Press, 2017.

Application of approximation rule



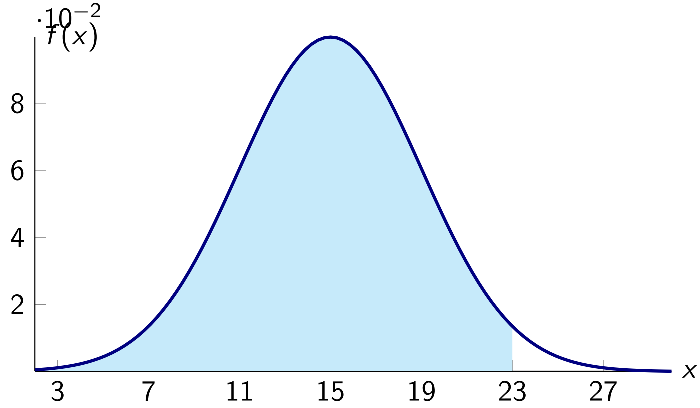
- Let X be the height of a male in inches.
- A height of 82 inches is two standard deviations above the mean. Hence area to the right of 82 inches is approx. 0.025 Hence, $P(X < 82) = 1 - 0.024 = 0.976$

Example: $X \sim N(15, 4)$



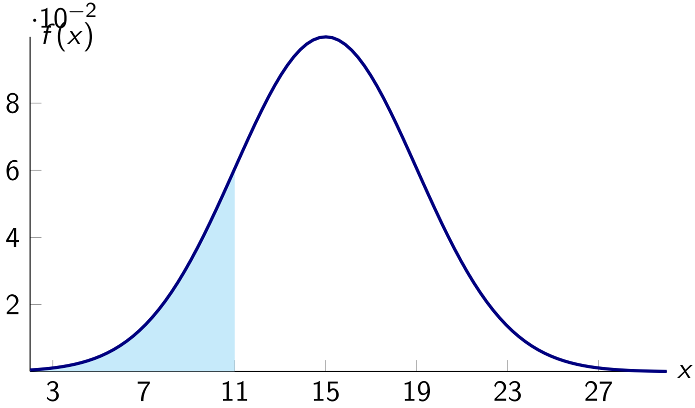
The probability that X is between 11 and 19 is approximately = 0.68

Example: $X \sim N(15, 4)$



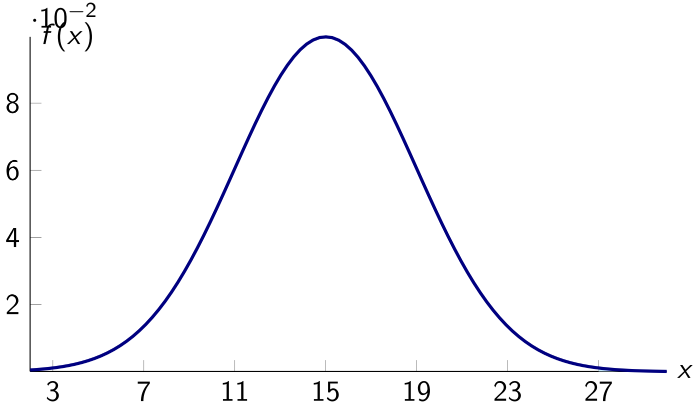
The probability that X is is less than 23 is approximately $= 0.975$

Example: $X \sim N(15, 4)$



The probability that X is less than 11 is approximately $= 0.16$

Example: $X \sim N(15, 4)$



The probability that X is greater than 27 is approximately $= 0$

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is between 7 and 19 is 0.95, then the expected value of X is approximately 13

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is greater than 16 is 0.975, then the expected value of X is approximately 22

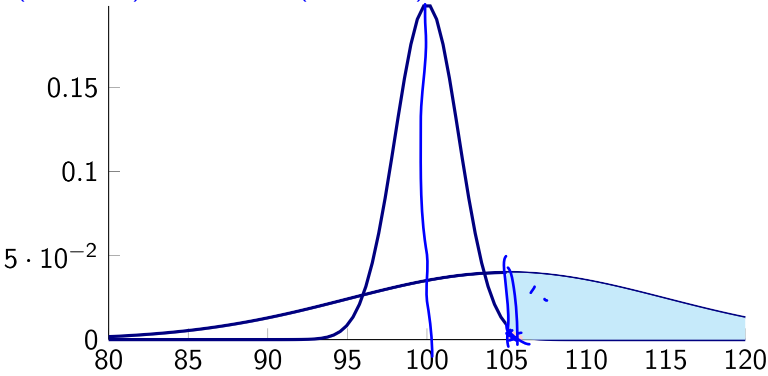
Example: Finding σ given probability and μ

Variable X is a normal random variable with expected value 100. If the probability that X is greater than 130 is 0.025, then the standard deviation of X is approximately 15

Comparing two distributions

If X is normal with expected value 100 and standard deviation 2, and Y is normal with expected value 105 and standard deviation 10, is X or is Y more likely to exceed 105?

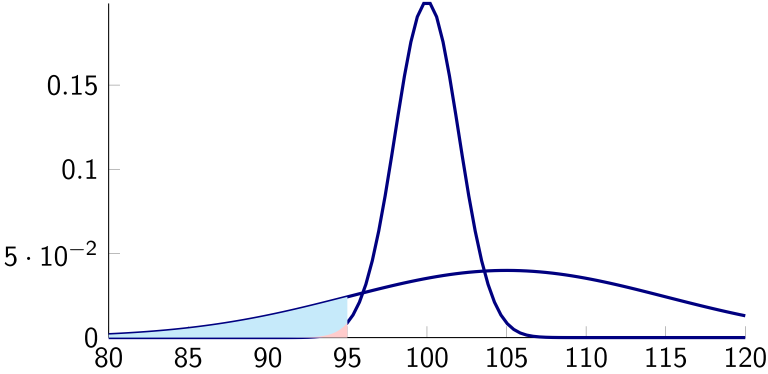
1. $P(X > 105) \approx 0.0062, P(Y > 105) = 0.5$



Comparing two distributions

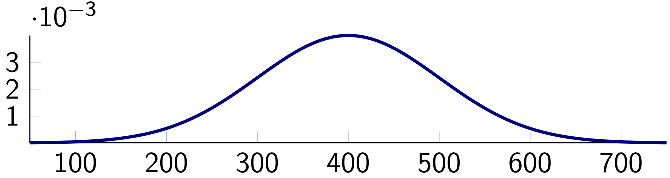
If X is normal with expected value 100 and standard deviation 2, and Y is normal with expected value 105 and standard deviation 10, is X or is Y more likely to be less than 95

1. $P(X < 95) \approx 0.0062, P(Y < 95) \approx 0.16$



Example

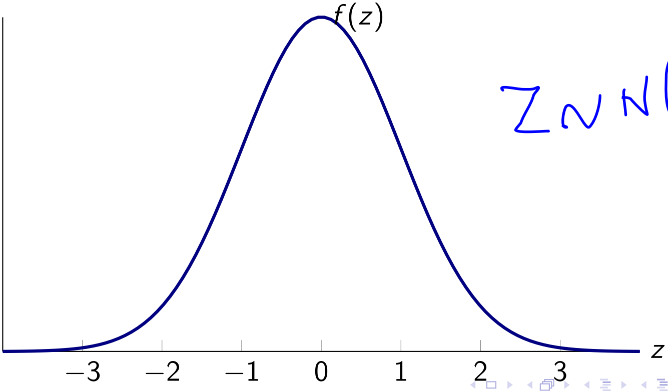
The scores on a particular job aptitude test are normal with expected value 400 and standard deviation 100. If a company will consider only those applicants scoring in the top 5 percent, determine whether it should consider one whose score is



1. 400 NO
2. 450 NO
3. 500 NO
4. 600 YES

Standard Normal Distribution: $\mu = 0$ and $\sigma = 1$

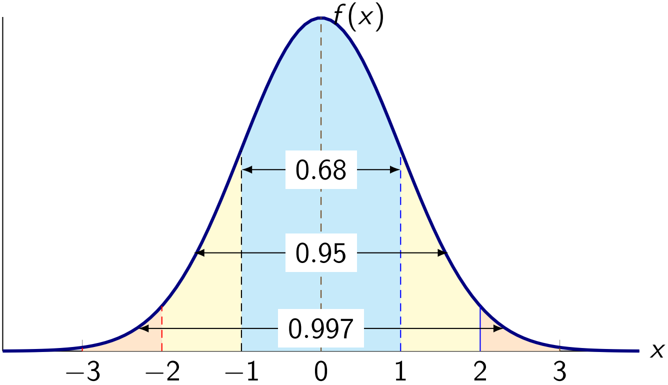
- A normal random variable having mean value 0 and standard deviation 1 is called a standard normal random variable, and its density curve is called the standard normal curve.
- We represent a standard normal random variable by Z.



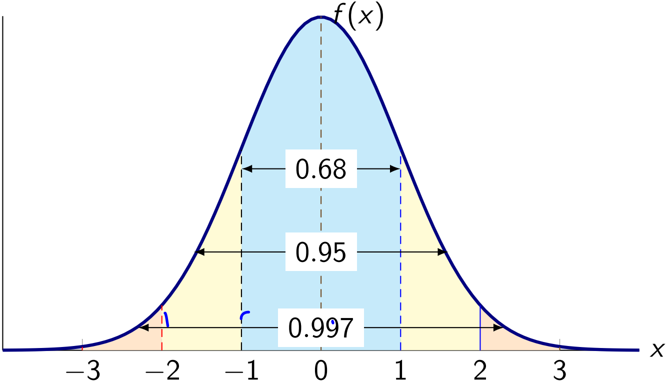
$Z \sim N(0, 1)$

Approximation rule for standard normal

- Between -1 and $+1$ with approx. probability 0.68
- Between -2 and $+2$ with approx. probability 0.95
- Between -3 and $+3$ with approx. probability 0.997

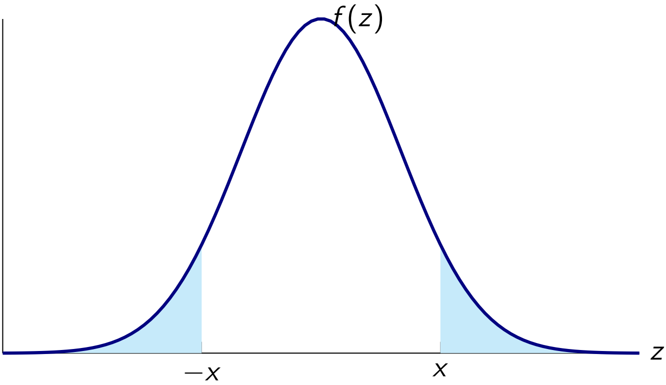


Application of approximate rule to standard normal probabilities



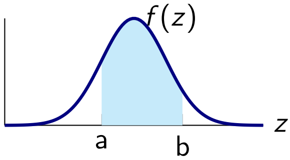
1. $P(-2 < Z < 2) = 0.95$
2. $P(Z > -1) = 0.84$
3. $P(Z > 1) = 0.16$
4. $P(Z > 3) = 0$
5. $P(Z < 2) = 0.975$

Property 1: $P(Z < -x) = 1 - P(Z < x)$

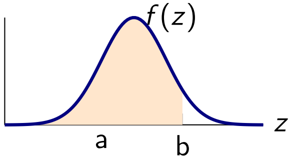


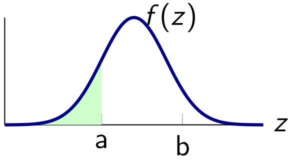
- $P(Z < -x) = P(Z > x) = 1 - P(Z < x)$

Property 2: $P(a < Z < b) = P(Z < b) - P(Z < a)$



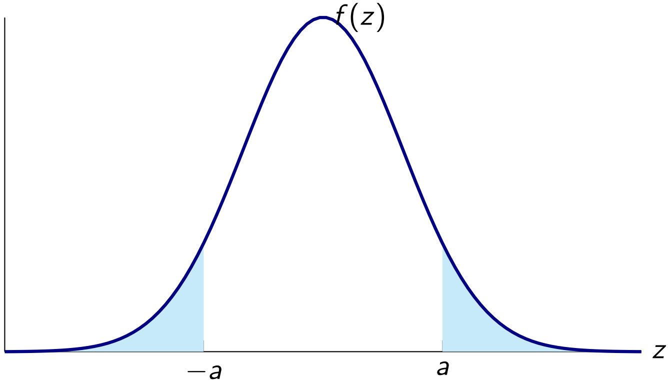
This can be expressed as difference of the areas:





- $P(a < Z < b) = P(Z < b) - P(Z < a)$

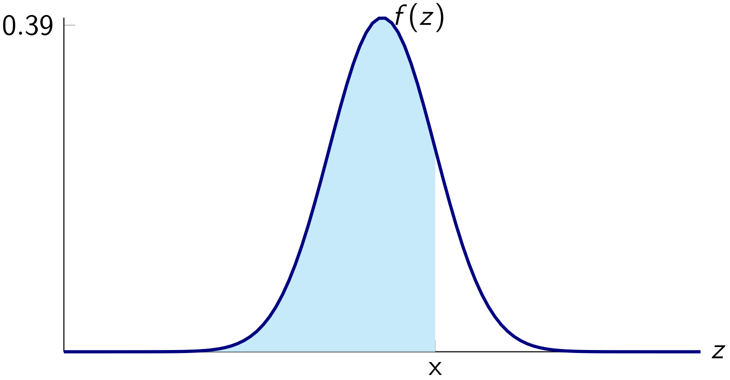
Property 3: $P(|Z| > a) = 2(1 - P(Z < a))$



- $P(|Z| > a) = P(Z > a) + P(Z < -a) = 2P(Z > a) = 2(1 - P(Z < a))$

Standard Normal table

- For each nonnegative value of x, the table specifies the probability that Z is less than x, i.e. $P(Z \leq x)$.



Portion of Standard Normal table

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
0	$\boxed{0.5}$	0.504	0.508	0.512	0.516	0.5199	0.5239
0.1	0.5398	0.5438	$\boxed{0.5478}$	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
0.8	0.7881	0.791	0.7939	0.7967	$\boxed{0.7995}$	0.8023	0.8051
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962

Standard Normal table

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.648	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.719	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.791	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.937	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.975	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.983	0.9834	0.9838	0.9842	0.9846	0.985	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.992	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.996	0.9961	0.9962	0.9963	0.9964

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

2. $P(Z > 1.1) = 0.1357$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	$\boxed{0.8643}$	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883

3. $P(0 < Z < 2) = 0.4772$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	$\boxed{0.8849}$	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.901

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$$P(-0.9 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.9) =$$
$$P(Z < 1.2) - P(Z > 0.9) =$$
$$P(Z < 1.2) + P(Z < 0.9) - 1 = 0.8849 + 0.8159 - 1 =$$
$$0.7008$$

Using Normal tables to compute probabilities

5. $P(Z > -1.96) = 0.9750$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

6. $P(Z < -0.72) = 0.2358$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.7	0.758	0.7611	$\boxed{0.7642}$	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852

Using Normal tables to compute probabilities

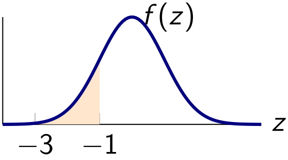
7. $P(|Z| < 1.64) = 0.899$
8. $P(|Z| > 1.20) = 0.2302$
9. $P(-2.2 < Z < 1.2) = 0.8710$

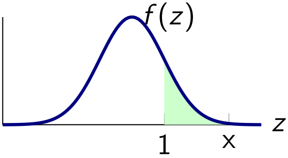
z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.2	$\boxed{0.8849}$	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.6	0.9452	0.9463	0.9474	0.9484	$\boxed{0.9495}$	0.9505	0.9515	0.9525	0.9535	0.9545
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

Using Normal tables to compute unknown x

Find the value of x:

$$P(-3 < Z < -1) = P(1 < Z < x)$$





solution: $x = 3$

Examples

Find value of x

1. $P(Z > x) = 0.05$, $x = 1.65$
2. $P(Z > x) = 0.005$, $x = 2.58$
3. $P(|Z| < x) = 0.75$, $2P(Z < x) = 1.75; x = 1.15$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	$\boxed{0.8749}$	0.877	0.879	0.881	0.883
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	$\boxed{0.9505}$	0.9515	0.9525	0.9535	0.9545
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	$\boxed{0.9951}$	0.9952

Standardizing to Find Normal Probabilities

$X \sim N(\mu, \sigma)$

$Z \sim N(0,1)$

- Start by converting x into a z-score

$$Z = \frac{X - \mu}{\sigma}$$

The value of the standardized variable tells us how many standard deviations the original variable is from its mean.

- We can compute any probability statement about X by writing an equivalent statement in terms of Z

$$P(X < a) = P\left(\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right) = P\left(Z < \frac{a - \mu}{\sigma}\right)$$

Example: Distribution of IQ scores

Suppose IQ examination scores for students of a particular school are normally distributed with mean value 140 and standard deviation 10.

1. What is the probability a randomly chosen student has a score greater than 130?

$$P(X > 130) = P(Z > -1) = 0.84$$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
1	$\boxed{0.8413}$	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554

2. What is the probability a randomly chosen student has a score between 120 and 150?

$$P(110 < X < 150) = P(-2 < Z < 1) = 0.8185$$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

1. $P(|X - \mu| > \sigma) = P(|Z| > 1) = 0.3174$
2. $P(|X - \mu| > 2\sigma) = P(|Z| > 2) = 0.0456$
3. $P(|X - \mu| > 3\sigma) = P(|Z| > 3) = 0.0026$

- The statement $|X - \mu| > a\sigma$ is, in terms of the standardized variable Z, is equivalent to the statement $|Z| > a$

Example: Transforming normal to Standard normal

X is normal with mean 10 and standard deviation 3, find

1. $P(X > 12) = P(Z > 0.67) = 0.2514$
2. $P(X < 13) = P(Z < 1) = 0.8413$
3. $P(8 < X < 11) = P(-0.67 < Z < 0.33) = 0.3779$
4. $P(X > 7) = P(Z < 1) = 0.8413$
5. $P(|X - 10| > 5) = P(|Z| > 1.67) = 0.095$
6. $P(X > 10) = P(Z > 0) = 0.5$

Example: Exam scores

The scores on an entrance exam were normally distributed with mean 500 and standard deviation 90.
Let X equal to the score on the scholastic test.

1. If your score was 700, by how many standard deviations did it exceed the average score?
   $Z = \frac{700-500}{90} = 2.22$. Thus, the score of 700 is 2.22 standard deviations above the mean of 500

2. What percentage of examination takers received a higher score than you did?

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2.2	0.9861	0.9864	$\boxed{0.9868}$	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

   $P(X > 700) = P(Z > 2.22) = 0.0132$. Thus, the percentage who scored higher than 700 is 1.32%

Example: lifetime of automobile tires

The life of a certain automobile tire is normally distributed with mean 35,000 miles and standard deviation 5000 miles.
Let X be the life of the tire in thousands of miles.

1. What proportion of such tires last between 30,000 and 40,000 miles? $P(30 < X < 40) = P(-1 < Z < 1) = 0.6826$
2. What proportion of such tires last over 40,000 miles? $P(X > 40) = P(Z > 1) = 0.1587$
3. What proportion last over 50,000 miles? $P(X > 50) = P(Z > 3) = 0.0013$

Example: Pulse rate

The pulse rate of young adults is normally distributed with a mean of 72 beats per minute and a standard deviation of 9.5 beats per minute. If the requirements for the military rule out anyone whose rate is over 95 beats per minute, what percentage of the population of young adults does not meet this standard?

- Let X be pulse rate of young adults
- $P(X > 95); X \sim N(72, 9.5)$
- $P(X > 95) = P(Z > 2.42) = 0.0078$
- Thus 0.78% of the population of young adults do not meet the military standard.

Example: Manufacturing bolts

The bolts produced by a manufacturer are specified to be between 1.09 and 1.11 inches in diameter. If the production process results in the diameter of bolts being a normal random variable with mean 1.10 inches and standard deviation 0.005 inch, what percentage of bolts do not meet the specifications?

- Let X be diameter of the bolts in inches.
- $X \sim N(1.1, 0.005)$
- $P(1.09 < X < 1.11) = P(-2 < Z < 2) = 0.9544$
- Thus the proportion of bolts that do not meet the specifications is $(1 - 0.9544) \times 100\% = 4.56\%$

Example: Junking car battery

You are planning on junking your old car after it runs an additional 20,000 miles. The battery on this car has just failed, and you must decide which of two types of batteries, costing the same amount, to purchase. After some research you have discovered that the lifetime of the first battery is normally distributed with mean life 24,000 and standard deviation 6000 miles, and the lifetime of the second battery is normally distributed with mean life 22,000 and standard deviation 2000 miles.

Example: Junking car battery-solution

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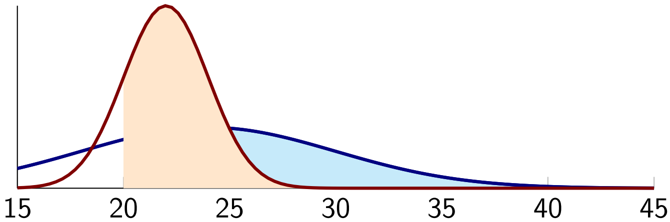
- If all you care about is that the battery purchased lasts at least 20,000 miles, which one should you buy?
- Let X =lifetime of first battery and let Y = lifetime of second battery.
- $P(X > 20) = 0.7486; P(Y > 20) = 0.8413$
- Since second battery has higher probability of lasting beyond 20,000 miles, purchase the second battery
- Would your answer change if you wanted the battery to last 21,000 miles?


15
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Section summary

- Standardizing a normal random variable
- Applications.

Additive Property

Result

The sum of independent normal random variables is also a normal random variable. Suppose X and Y are independent normal random variables with respective parameters μ_X, σ_X and μ_Y, σ_Y, then $X + Y$ also will be normal with mean $\mu_X + \mu_Y$ and standard deviation $\sqrt{\sigma_X^2 + \sigma_Y^2}$.

Application of additive property: lifetime of two bulbs

Suppose the amount of time a light bulb works before burning out is a normal random variable with mean 400 hours and standard deviation 40 hours. If an individual purchases two such bulbs, one of which will be used as a spare to replace the other when it burns out, what is the probability that the total life of the bulbs will exceed 750 hours?

- $P(X+Y) > 750 = ?$
- $X + Y \sim N(800, \sqrt{3200})$
- $P(X+Y > 750) = P(Z < 0.884) = 0.81$
- Therefore, there is an 81 percent chance that the total life of the two bulbs exceeds 750 hours.

Application of additive property: Computing rainfall over two years

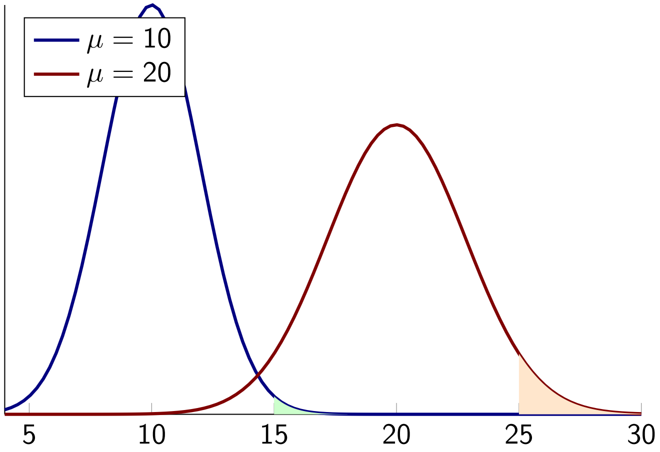
The annual rainfall in Guwahati, is normally distributed with mean 67 inches and standard deviation 8 inches.
Let X be annual rainfall in inches in Guwahati.

1. What is the probability that this year's rainfall exceeds 70 inches?
   $P(X > 70) = P(Z > 0.375) = 0.3538$
2. What is the probability that the sum of the next 2 years? rainfall exceeds 140 inches?
   let Y be sum of next two years rainfall. $Y \sim N(134, 11.313)$.
   $P(Y > 140) = P(Z > 0.53) = 0.297$

Comparing distributions

Let X_1 and X_2 be independent normal random variables, each having mean 10 and variance 4. Which probability is larger

1. $P(X_1 > 15)$
2. $P(X_1 + X_2 > 25)$
3. $P(X_1 + X_2 > 30)$


$\mu = 10$
$\mu = 20$
5
10
15
20
25
30

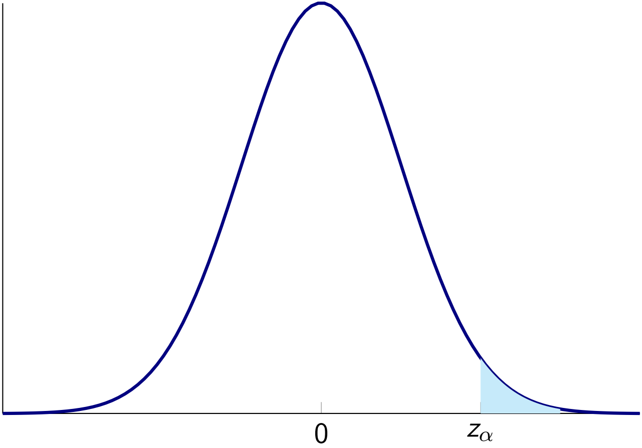
Section summary

1. Additive property of normal random variables and its applications.

Percentiles

- The previous problems ask us to find the probability associated with being larger or smaller than some value.
- Some problems require working in the opposite direction that starts with the probability
- For any α between 0 and 1, we define z_α to be that value for which
$$P(Z > z_\alpha) = \alpha$$
- z_α is the $100(1-\alpha)$ percentile of the standard normal distribution.

Percentile



$$P(Z > z_\alpha) = \alpha$$

$z_{0.025}$

- What is $z_{0.025}$? $P(Z < z_{0.025}) = 1 - P(Z > z_{0.025}) = 0.975$.

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

- Search for entry 0.975 in table and find value of x corresponding to this value. We see that $z_{0.025} = 1.96$
- That is, 2.5 percent of the time a standard normal variable will exceed 1.96.
- Since 97.5 percent of the time a standard normal random variable will be less than 1.96, we say that 1.96 is the 97.5 percentile of the standard normal distribution.
- In general, since $100(1-\alpha)$ percent of the time a standard normal random variable will be less than z_α, we call z_α the $100(1-\alpha)$ percentile of the standard normal distribution.

$z_{0.05}$

- What is $z_{0.05}$?
- For the value 0.95, we do not find this exact value. Rather, we see that $P(Z < 1.64) = 0.9495$ and $P(Z < 1.65) = 0.9505$
- Therefore, it would seem that $z_{0.05}$ lies roughly halfway between 1.64 and 1.65, and so we could approximate it by 1.645.

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.0**
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9

- The values of $z_{0.10}, z_{0.05}, z_{0.025}, z_{0.01}$, and $z_{0.005}$, are of particular importance in statistics. Their values are as follows:

$$z_{0.10} = 1.282, z_{0.025} = 1.96, z_{0.005} = 2.576$$

$$z_{0.05} = 1.645, z_{0.01} = 2.326$$

Percentiles of any normal random variable

- To obtain the percentiles of any normal random variable X, convert X to the standard normal Z.
- For instance suppose we want to find the value of a for which $P(X < a) = 0.95$, when X is normal with mean 40 and standard deviation 5.
- $$\begin{aligned} 0.95 &= P(X < a) \\ &= P\left(\frac{X-40}{5} < \frac{a-40}{5}\right) \\ &= P\left(Z < \frac{a-40}{5}\right) \end{aligned}$$
- But $P(Z < z_{0.05}) = 0.95$, so we obtain

$$\frac{a-40}{5} = z_{0.05} = 1.645$$

- Hence, the desired value of $a = 5(1.645) + 40 = 48.225$

Application: IQ scores

An IQ test produces scores that are normally distributed with mean value 100 and standard deviation 14.2. What is the range of scored in the top 1 percent of all scores?

- We want to find the value of a for which $P(X > a) = 0.01$ when X is normal with mean 100 and standard deviation 14.2
- Now $P(X > a) = P\left(\frac{X-100}{14.2} > \frac{a-100}{14.2}\right)$
- $\frac{a-100}{14.2} = z_{0.01} = 2.33$
- $x = 14(2.33) + 100 = 133.086$
- That is, the top 1 percent consists of all those having scores above 134.

In general

Let X be a normal random variable with mean μ and standard deviation σ

$$P(X > \mu + \sigma z_\alpha) = \alpha$$